

Constraint-Based Problem Solving

Problem Statements — Questions 1, 2 and 3

Question 1: Hospital Shift Scheduling (CSP)

A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts — Morning, Afternoon, and Night — such that all of the following constraints are satisfied simultaneously:

- (i) D1 cannot work the Night shift.
- (ii) D2 must work before D3, where the shift ordering is Morning < Afternoon < Night.
- (iii) Only one doctor may be assigned per shift (all assignments must be distinct).
- (iv) D3 cannot work the Morning shift.
- (v) Every doctor must be assigned exactly one shift (no doctor is left unassigned).

Task: Apply Backtracking Search to find a valid assignment, showing all intermediate steps and constraint checks. State the final valid schedule.

Question 2: Robot Grid Navigation (BFS)

A robot operates in a 5×5 grid environment and must move from a Start cell (S) to a Goal cell (G), while avoiding obstacles, subject to the following constraints:

- (i) Allowed moves: Up, Down, Left, Right (no diagonal movement).
- (ii) Cost of each move = 1 (uniform step cost).
- (iii) The robot cannot pass through cells marked as obstacles.
- (iv) Manhattan Distance is used as the heuristic reference to judge optimality of the path found.

Task: Using the above constraints and the BFS Search Algorithm, show step-by-step node expansion and priority-queue updates, and determine the optimal path and cost.

Question 3: Autonomous Rescue Robot (Online / UCS Search)

An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information. The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move Up, Down, Left, Right.
- Some cells are blocked (obstacles) and cannot be traversed.
- The robot has limited sensing capability (can only observe adjacent cells).
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2.
- The robot must minimize total cost while ensuring safe navigation.
- The robot must update its knowledge dynamically (online search scenario).

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as: Intelligent agents, Rationality, Environment type.
- d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.